

Methodology: Implementation Details

This section provides pseudocode for major algorithms (sec:pseudocode), configuration parameters (sec:config-params), framework API reference (sec:framework-api), deployment considerations (sec:deployment-checklist), and integration examples (sec:integration-examples).

Cross-Reference Note: *All algorithms in this section implement the formal definitions and theorems from Part 1 of this series. We cite specific theorems and definitions using the notation “(Part 1, Theorem X.Y)” to enable readers to trace implementations to their theoretical foundations.*

Pseudocode for Major Algorithms

Algorithm 1: Cognitive Firewall

The cognitive firewall classifies incoming messages using a multi-stage detection pipeline. This algorithm implements the formal Cognitive Firewall definition from Part 1, Section 5.1, which specifies the three-stage filtering ($F_{sig} \rightarrow F_{sem} \rightarrow F_{anom}$) with combined threat scoring (Part 1, Definition 5.1).